



June 17, 2026

Dear Admissions Committee,

I am writing to express my enthusiastic interest in pursuing a PhD at the University of Southampton under the project "Shadow Tomography for In-Context Quantum Machine Learning". I am impressed by the university's cutting-edge research in artificial intelligence and its applications in quantum computing.

As an AI Data Science Engineer with a strong computational background, I believe my skills align perfectly with the requirements of this PhD offer. My most relevant achievement that resonates with this project is my development of *MedicalMambaTS—a linear-time state space model for sub-millisecond bedside inference. This achievement enables speed decision-making, which is crucial in quantum computing where encoding and extracting information from*

Moreover, my work on *PI\_DeepONet—a 0D Windkessel Physics-Informed Operator Network using PyTorch* showcases my proficiency in handling physical constraints and complex systems. In this context, I believe I can *inspire quantum circuit architectures that incorporate physical constraints from the observation of "in-context learning" in large language models.*

I am particularly drawn to this project because it leverages ideas from observation of "in-context learning" in large language models into quantum computing, which aligns with my passion for exploring the intersection of AI and physics. I am confident that my skills and experience will enable me to make a meaningful contribution to this project and produce high-quality research.

Thank you for considering my application. I look forward to discussing my qualifications further.

Sincerely,

Ismail Aissaoui